

ASSIGNMENT 3 - ADVERSARIAL ROBUST IMAGE CLASSIFICATION

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1 INTRODUCTION

The objective of this task is to train an image classifier capable of maintaining high predictive performance on both clean data and adversarially perturbed inputs. Machine learning models, particularly deep neural networks, are highly susceptible to adversarial examples—inputs intentionally designed with subtle, often imperceptible noise to force incorrect predictions. Developing defenses against these vulnerabilities is a critical aspect of trustworthy machine learning. Our goal is to maximize a unified evaluation metric that equally weights clean and robust accuracy, ensuring the classifier does not merely overfit to a specific attack but demonstrates genuine, generalized robustness across varied threat models.

2 METHODOLOGY AND STRUCTURAL CHOICES

2.1 ARCHITECTURE

We utilize a ResNet18 architecture, modified at the final fully connected layer to output logits for the 9 target classes. While deeper models such as ResNet34 and ResNet50 were considered, ResNet18 was selected due to its optimal balance between representational capacity and computational efficiency. Since adversarial training via PGD is computationally expensive, larger models significantly slow down training without proportional gains in robustness for low-resolution ($3 \times 32 \times 32$) images. Additionally, the smaller parameter count of ResNet18 acts as an implicit regularizer, reducing overfitting risks under adversarial training.

2.2 ADVERSARIAL ATTACK FORMULATION

Adversarial examples are generated dynamically during training using Projected Gradient Descent (PGD). The attack is configured with 10 iterative steps, a maximum perturbation bound of $\epsilon = 8/255$, and a step size of $\alpha = 2/255$.

2.3 MIXED-BATCH TRAINING STRATEGY

To balance clean and robust accuracy, we adopt a 50/50 mixed-batch training strategy. Each training batch contains an equal split of clean (augmented) images and their adversarially perturbed counterparts. This prevents catastrophic forgetting of natural features while maintaining robustness against adversarial perturbations, ensuring that clean accuracy remains above the required threshold.

2.4 HYPERPARAMETERS

The model is optimized using Stochastic Gradient Descent (SGD) with a learning rate of 0.1, Nesterov momentum of 0.9, and weight decay of 5×10^{-4} . A Cosine Annealing learning rate scheduler is applied over 100 epochs to stabilize convergence toward a robust minimum. Additionally, label smoothing (0.1) is used in the Cross-Entropy loss to reduce overconfidence on adversarial samples.

2.5 KEY RESULTS

Initial experiments using only adversarial batches led to severe overfitting to PGD-specific perturbations, causing a drop in clean accuracy below acceptable limits. Introducing the 50/50 mixed-batch strategy stabilized training and significantly improved generalization.

After 100 epochs, the model achieved:

- Clean accuracy: **89.1%**
- Robust accuracy: **37.6%**

On the public leaderboard, the model achieved a final unified score of **0.649957**, demonstrating strong generalization to unseen adversarial attack configurations.

3 CONCLUSION

The presence of adversarial vulnerabilities highlights a critical gap between empirical model performance and real-world deployment safety. In safety-critical systems such as autonomous driving, medical diagnosis, and biometric authentication, even imperceptible input perturbations can lead to catastrophic failures. This makes adversarial robustness not just an academic concern but a practical necessity.

By incorporating adversarial training using PGD and a balanced mixed-batch strategy, we shift from standard empirical risk minimization toward worst-case robust optimization. This paradigm is essential for building reliable machine learning systems that can be safely deployed in real-world environments where adversaries may actively attempt to manipulate model behavior.

CODE REPOSITORY

<https://github.com/shreyakolhapure9130-byte/Adversarial-Robust-Image-Classification>

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